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ITC-C508 | LAB EXERCISE WW-M2 · RESEARCH PROPOSAL

Evaluating a RAG-Based Chatbot for User-Selected Undergraduate Curriculum Comparisons at José Rizal University

DecisionThree

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The Problem: Too Much Information, Too Little Clarity

1

Fragmented Sources

Grade 12 students comparing JRU programs must dig through multiple documents just to compare subjects, units, and practical requirements.

2

Complexity Multiplies

The more programs compared simultaneously, the harder it becomes to identify meaningful differences — and make an informed choice.

3

Unvalidated Assumption

We don't yet know if this is a real pain point for JRU applicants. A short needs-validation study is the essential first step.

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SDG 4: QUALITY EDUCATION

Equal Access Starts With Clear Information

Target 4.3 calls for equal access to quality tertiary education. Students need transparent, comparable program information *before* they enroll — not after.

Research shows clear program data can shift undecided students' choices **(Ersoy & Speer, 2025)** — though JRU-specific evidence is still needed.

A RAG chatbot pulls approved curriculum docs, surfaces differences, and cites sources — turning fragmented PDFs into structured, trustworthy answers.

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Narrowing the Research Focus

From a broad SDG concern to a testable, bounded study — each level sharpens the question.

1	**Broad**	Access to tertiary program info
2	**SDG Issue**	Hard to consolidate curriculum info
3	**Users**	Graduating Grade 12 students
4	**Narrow Problem**	Comparing subjects, units, and requirements across JRU CSE programs
5	**Chatbot Focus**	Comparative queries across 2 or more CSE programs

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Scope & Boundaries

IN SCOPE

- All 5 CSE programs: BSCpE, BSEcE, BSIT, BSIT-BA, BSEMC
- BSEMC majors (Digital Animation, Game Dev) — compared at major level
- Subjects, units, and practical requirements
- Comparative queries only

OUT OF SCOPE

- Grade prediction or academic advising
- “Best fit” recommendations
- Admissions processing or enrollment decisions
- Programs outside the CSE college

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CORE RESEARCH QUESTION

As students compare more JRU programs at once, how much do answer correctness, groundedness, and completeness drop off — if at all?

Test Design

Comparisons are tested at 2-program, 3-program, and full-set sizes to measure retrieval quality under increasing context load.

Why This Question Matters

Unlike a simple RAG vs. no-RAG test, this outcome is genuinely unknown. It's grounded in Liu et al.'s “Lost in the Middle” findings on long-context retrieval degradation in LLMs.

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Data Sources

01 **Primary Sources**

5 official CSE curriculum PDFs published by JRU (April 2024)

02 **Supporting Sources**

JRU CSE program listing page — confirms the full program set

03 **Supplementary Sources**

Subject descriptors and internship documents where curriculum PDFs lack detail

04 **Source Logging**

All sources tracked by program, major, curriculum version, and verification date

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Evaluation Plan

A rigorous, faculty-reviewed test set ensures findings are credible and reproducible.

~40

Test Questions

30 answerable, 5 partial, 5 out-of-scope

3

Comparison Sizes

2-program, 3-program, and full-set

4

Evaluation Metrics

Correctness, groundedness, completeness, and citation support

2

Independent Scorers

Faculty-reviewed answer key with dual scoring for reliability

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Proposed Model & Manageability

Qwen3-8B (Apache 2.0, open-weight) is the preliminary model choice — not final. Its compact size suits a single-task, single-university, fixed-corpus study.

This is a proposal — no data has been collected, and nothing has been built yet.

Document Access

Official curriculum PDFs must be secured

Needs Validation

Short survey to confirm the problem exists at JRU

Reviewer Availability

Faculty reviewers for answer-key validation

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